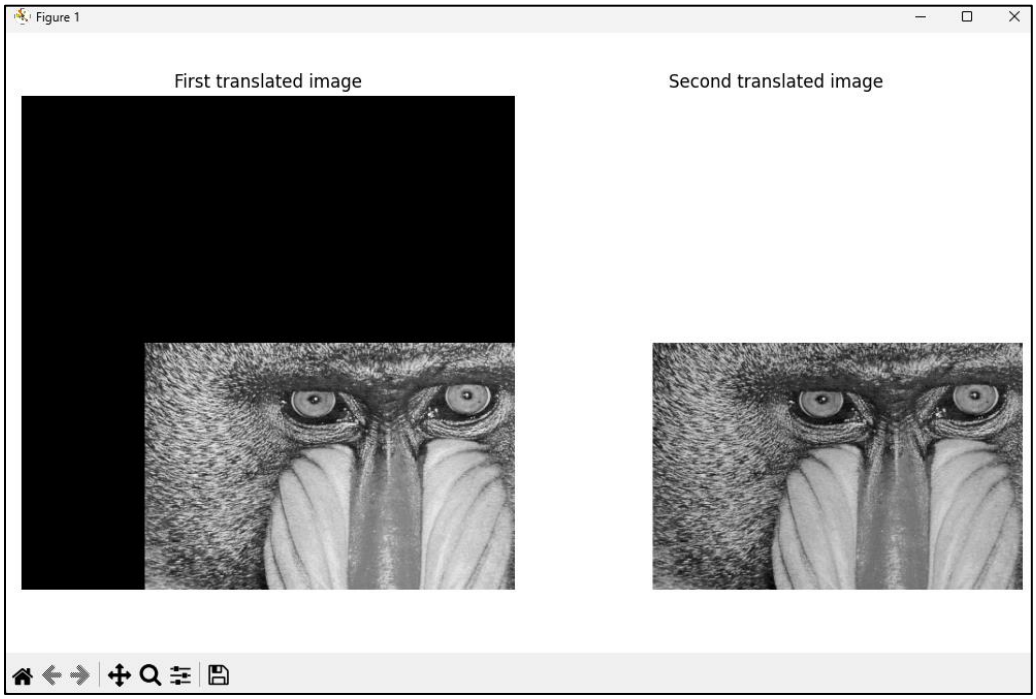
# Exercise for Assignment 3 (Week 7)

Affine transformation: If interpolation is needed, use nearest-neighbor interpolation.

Write a function $image\ Translate4e(f, tx, ty, mode)$ for performing image translation, where $f$ is a grayscale image and $tx$ and $ty$ are translation factors (can be any real number) in the $x$ (horizontal) direction and $y$ (vertical) directions. The output image should be of the same size as the input, and its background in the area vacated by the translated image is to be determined by *mode*. If $mode = 'black'$, the background should be black (this is the default if *mode* is not included in the function call). If $mode = 'white'$, the background should be white. Test your function by translating image "BaboonGray.png" by half its height in the positive vertical direction and by one-fourth of its width in the positive horizontal direction. Test both values of *mode*. **You are NOT allowed to use any existing affine-transformation functions, including but not limited to OpenCV helpers such as cv2.warpAffine, cv2.getAffineTransform, or cv2.remap. Implement translations/affine operations manually (e.g., with direct pixel/index mapping using NumPy) instead of calling OpenCV affine utilities.**

*Hint*: The translation equation is:

$$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix}$$




## Requirements:

- Only the following packages are permitted: OpenCV-Python, Matplotlib, NumPy
- Use Python version 3.10+